Answer **one** question from this section in the spaces provided.

8 **(a)** **(i)** Define *gravitational field strength*.

.....

.....

..... [1]

(ii) Derive, from Newton's Law of Gravitation and **(a)(i)**, the equation for the gravitational field strength of a point mass of mass M_1, placed in a gravitational field generated by a mass M_2 and at a distance of r from M_1. [2]

(b) Information related to the Earth and the Moon is given below:

$$\frac{\text{Radius of Earth}}{\text{Radius of Moon}} = 3.7$$

$$\frac{\text{Mass of Earth}}{\text{Mass of Moon}} = 81$$

The center-to-center distance of the Moon from the Earth is 3.84×10^8 m and the gravitational field strength due to the Earth at its surface is 9.8 N kg^{-1}.

(i) Using these data, calculate the gravitational field strength due to the Moon at its surface.

gravitational field strength = ... N kg^{-1} [3]

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(ii) There is a point on the line between the Earth and the Moon at which their combined gravitational field strength is zero.

Calculate the distance between this point and the centre of the Earth.

distance = …… m [2]

(iii) The Moon orbits around the Earth with a period of 27.3 days.

1. Calculate the angular speed of the Moon.

angular speed = …….. rad s^{-1} [1]

2. Calculate the mass of the Earth.

mass = …… kg [2]

3. Determine the gravitational force between the Earth and the Moon.

gravitational force = …….. N [2]

4. Tidal action on the Earth's surface causes the radius of the orbit of the Moon to increase by 4.0 cm each year. Using your answer in **(b)(iii)(3)**, determine the change, in one year, of the gravitational potential energy of the Moon.

change in potential energy = …….. J [2]

(c) **(i)** Explain, by considering the respective field forces, why gravitational potential energy is negative whereas electric potential energy can be positive or negative.

……

……

……..[1]

(ii) The Earth may be assumed to be an isolated sphere of radius 6.4×10^3 km with its mass of 6.0×10^{24} kg concentrated at its centre. A 2.0 kg mass is projected vertically from the surface of the Earth so that it reaches a maximum altitude of 1.3×10^4 km.

Calculate, for this mass,

1. the change in gravitational potential energy

change in gravitational potential energy = …….. J [2]

2. the speed of projection from the Earth's surface, assuming air resistance is negligible.

speed = …….. m s^{-1} [2]